
CMS Physics Analysis Summary

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Search for high mass resonances decaying into W^+W^- in the dileptonic final state with 138 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

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Abstract

A search for high mass resonances decaying into a pair of W bosons is performed. The analysis considers the fully leptonic final state ($e\mu$, $\mu\mu$, ee). New techniques are implemented in the analysis to improve the sensitivity of the search, especially in the very high mass range. The search is performed in a mass range from 115 GeV to 5 TeV, and for various width hypotheses. Both the gluon-gluon-fusion and vector-boson-fusion signal production processes are considered. Interference effects between signal and background are also taken into account. The results are presented as 95% confidence level upper limits on the product of the cross section and branching ratio on the production of a high mass resonance, and exclusion limits are derived on various two-higgs-doublet models and minimal supersymmetric standard model benchmark scenarios.

1 Introduction

In 2012 a new particle was discovered by the ATLAS and CMS experiments at the LHC [1, 2]. This new particle, with a mass of about 125 GeV, is consistent with the standard model (SM) Higgs boson. With this discovery the Higgs mechanism has been validated, solving one of the open mysteries in particle physics. There are however many more open questions in particle physics, such as the incorporation of gravity in the SM theory, or the existence of dark matter. Many theories attempting to provide answers to such questions predict the existence of new particles. These can be, for example, additional Higgs bosons with properties similar to those of the Higgs boson discovered in 2012, but with a larger mass. Finding evidence for such a particle would shed light on the missing pieces of the SM and help to complete our understanding of the universe.

This document presents a search for a high mass resonance decaying into a pair of W bosons, using the full data set recorded by CMS during the LHC Run 2. Data are from proton-proton collisions at a center of mass energy of $\sqrt{s} = 13$ TeV recorded between 2016 and 2018 and corresponding to an integrated luminosity of 138 fb^{-1} . The analysis considers the fully leptonic (or dileptonic) final state, where both W bosons decay into electrons or muons, along with neutrinos. A distinction is made between the opposite flavor (OF) final state, $e\mu$, and the same flavor (SF) final state, $\mu\mu$ and ee . The search is performed in a mass range from 115 GeV to 5 TeV, and for various width hypotheses of the signal. Event categorisation optimised for the gluon-gluon-fusion (ggF) and vector-boson-fusion (VBF) production mechanisms are used to increase the signal sensitivity.

We perform a signal interpretation in terms of a heavy Higgs boson with SM-like couplings and decays, motivated by models of beyond the SM physics in which the SM Higgs boson mixes with a heavy electroweak singlet, giving rise to an additional resonance at high mass with couplings similar to those of the SM Higgs boson [3]. Both the ggF and VBF signal production processes are considered, with a number of hypotheses for their relative contribution investigated. Interference between the resonance and the SM Higgs bosons, as well as with the non-resonant WW continuum, is taken into account. A number of interpretations based on two-higgs-doublet model (THDM) [4] formulations are also performed. The THDM introduces a second scalar doublet and is incorporated in supersymmetric models [5] and axion models [6]. Further interpretations are performed on minimal supersymmetric standard model (MSSM) benchmark scenarios [7, 8].

2 The CMS detector

The Compact Muon Solenoid (CMS) apparatus [9] is a multipurpose, nearly hermetic detector, designed to trigger on [10, 11] and identify electrons, muons, photons, and (charged and neutral) hadrons [12–15]. A global reconstruction “particle-flow” algorithm [16] combines the information provided by the all-silicon inner tracker and by the crystal electromagnetic and brass-scintillator hadron calorimeters, operating inside a 3.8 T superconducting solenoid, with data from gas-ionization muon detectors interleaved with the solenoid return yoke, to build τ leptons, jets, missing transverse momentum, and other physics objects [17–19].

3 Data and simulated samples

3.1 Data

This analysis is based on events that were triggered on one or two leptons, considering both electrons and muons, with transverse momenta (p_T) above a certain threshold. The exact thresholds depend on the trigger type and the year of data taking. Typical values of around 23 and 12 GeV were used for the leading (highest p_T) and subleading (second highest p_T) lepton for the dilepton triggers. The requirements on the lepton p_T are however tighter for events selected in this analysis, as discussed later.

3.2 Monte Carlo Simulations

Monte Carlo (MC) samples are used to model the expected signal for various mass hypotheses and the majority of background processes.

The ggF and VBF signal samples model the production and decay of a SM-like Higgs boson into a pair of W bosons, which then further decay into two charged leptons (e or μ) and two neutrinos. The signal samples are generated with POWHEG v2 [20–24] for masses of the Higgs boson between 115 GeV and 5000 GeV. The decay of the high mass resonance into two W bosons is modelled by the JHUGEN Monte Carlo program [25–29]. Different versions of JHUGEN were used in the production of different mass samples and for the running conditions corresponding to different years of data taking.

The backgrounds considered in the analysis include contributions from Drell-Yan (DY), processes involving top quarks, non-resonant di- and tri-boson states including WW production, as well as SM Higgs production. The estimation of these backgrounds is based on MC predictions from the different generators.

DY events are simulated using MADGRAPH5_AMC@NLO v2.4.2 [30]. Both the MLM [31] and FxFx [32] schemes to merge the matrix elements and the parton showers are used with MADGRAPH5_AMC@NLO depending on the sample. Most of the top quark background events are simulated with POWHEG v2. Events from the s-channel of single-top events are simulated with MADGRAPH5_AMC@NLO v2.4.2 with the MLM scheme. MADSPIN [33] is used in addition to POWHEG for single-top events in the t-channel. The quark-initiated WW events are simulated with POWHEG v2. The $gg \rightarrow WW$ sample specifically is produced by the MCFM v7.0 generator [34–37] when simulated with 2016 conditions. For 2017 and 2018 conditions, the sample is produced by MCFM v7.0.1 along with JHUGEN v7.0.11. Other WW background events are simulated with MADGRAPH5_AMC@NLO. Diboson events other than WW with a fully leptonic final state are simulated with POWHEG v2, and those where a boson decays hadronically are simulated with MADGRAPH5_AMC@NLO v2.4.2 with the FxFx scheme and MADSPIN. The latter, without MADSPIN, is also used for events simulating the production of a Z boson in association with a photon. For W bosons produced with a photon, MADGRAPH5_AMC@NLO v2.4.2 is used with the MLM scheme. SM Higgs boson events are simulated POWHEG v2.

For all samples, the detector response is simulated using a detailed description of the CMS detector, based on GEANT4 [38]. The event generator PYTHIA 8.2 [39] is used to model additional parton showers and interactions. In simulations for 2016 conditions, PYTHIA is used with the CUETP8M1 tune [40], which makes use of the PDF set NNPDF 3.0 [41]. For 2017 and 2018 conditions, the CP5 tune [42] is used instead, along with the complimentary NNPDF 3.1 PDF set [43].

4 Signal models

4.1 Signal width

The signal samples are reweighted using the matrix element likelihood analysis (MELA) package [28] to model different width scenarios, e.g. assuming a relative width based on the resonance mass. Weights are obtained for various relative widths between 0.1% and 10%. Unless specified otherwise, a relative width of 2% is the main scenario considered in this analysis.

For signal samples of resonance masses $m_H < 300$ GeV the signal distributions are not reweighted to a different width, because the signal samples are generated assuming the width of a SM-like Higgs boson at the corresponding mass. A 2% relative width is significantly larger, leading to ill-defined signal distributions because of the low statistical precision of off-shell events. The threshold of 300 GeV was chosen as the width of a SM-like Higgs boson of that mass corresponds to that of the 2% relative width scenario.

4.2 Interference

Since we consider scenarios where the high mass resonances have a non-zero width, the interference between the resonance and the SM Higgs bosons, as well as with the non-resonant WW continuum, becomes non-negligible. The contribution of the interference to the signal distribution, whether constructive or destructive, is determined by reweighting signal samples using the MELA framework. For signal masses $m_H < 300$ GeV, where no different width scenario is applied, interference effects are not considered, as their contributions for signals with very small widths are negligible.

4.3 Scenarios on production mechanisms

A parameter f_{VBF} , which is the fraction of the VBF production cross section with respect to the total cross section, is included in the signal model. The scenario with $f_{VBF} = 0$ ($f_{VBF} = 1$) considers only the ggF (VBF) production signal. The parameter may also be left floating in the final fit to the data. We note that the fraction of the VBF contribution is expected to increase for high mass resonances. In the final scenario considered f_{VBF} is set to the SM expectation using the cross sections provided by the LHC Higgs Cross Section Working Group [44].

4.4 THDM and MSSM benchmark scenarios

For this analysis we consider a number of different THDM scenarios. The expected cross sections and branching ratios of the heavy scalar H boson were determined as a function of the free parameters of the THDM given below. The ggF cross sections were calculated using SUSHi [45, 46], which includes next-to-leading order (NLO) quantum chromodynamic (QCD) corrections [47], next-to-next-to-leading order (NNLO) QCD corrections for the top quark contribution in the effective theory of a heavy top quark [48–50], and electroweak effects by light quarks [51, 52]. For the VBF cross sections we use the values from the LHC Higgs Cross Section Working Group, which are an approximation as they correspond to the SM-like Higgs boson scenario. To obtain the correct VBF cross sections for the heavier scalar Higgs boson H of the THDM, the cross sections are multiplied by $\cos^2(\beta - \alpha)$ [53], where α is the mixing angle between the two scalar Higgs bosons of the THDM, and β is defined from the vacuum expectation values of the two Higgs doublets as $\tan \beta = v_2/v_1$. The branching ratios are determined from 2HDMC [54, 55].

In the scenarios considered, the mass of the lighter scalar Higgs boson h is assumed to be $m_h = 125$ GeV and the masses of A and $H^\pm$ are assumed to be the same as for H. The mass m_H , along

with α and β , are the 3 free parameters of the model. THDM scenarios are therefore produced in a two-dimensional phase space of $\cos(\beta - \alpha)$ and $\tan \beta$ for several assumptions at fixed m_H , and as a function of m_H and $\tan \beta$ with the assumption $\cos(\beta - \alpha) = 0.1$. This is a reasonable assumption, as larger values are already excluded from different analyses [56, 57], and values smaller than 0.1 near the alignment limit result in a reduced sensitivity in this analysis, as the coupling of H to two vector bosons diminishes. The alignment limit is the case where the couplings of h become SM-like, which occurs at $\cos(\beta - \alpha) \rightarrow 0$. It is different from the decoupling limit, which is attained when $m_H \gg v$, where v is the vacuum expectation value of the SM.

In addition to the THDM scenarios, exclusion limits are also provided in the MSSM. The MSSM scenarios used in this analysis are provided by Refs. [7, 8], the baseline MSSM scenario is M_h^{125} . In this scenario the superparticles have a large enough mass that they barely influence MSSM Higgs boson processes. As a result, it resembles a THDM with MSSM-like couplings. The other scenarios studied in this analysis are:

- $M_h^{125}(\tilde{\tau})$: With respect to the baseline scenario (M_h^{125}), this scenario considers light staus.
- $M_h^{125}(\tilde{\chi})$: With respect to the baseline scenario, this scenario considers light neutralinos and charginos.
- $M_h^{125}(\text{alignment})$: This scenario considers alignment without decoupling. The lighter scalar h is the SM Higgs boson.
- $M_{h,\text{EFT}}^{125}$: With respect to the baseline scenario, this scenario is designed specifically for the low $\tan \beta$ region and is only defined up to $\tan \beta = 10$. The SUSY mass scale is adjusted throughout the m_A - $\tan \beta$ -plane to set the mass of the light CP-even Higgs boson h close to 125 GeV.
- $M_{h,\text{EFT}}^{125}(\tilde{\chi})$: This scenario is like $M_{h,\text{EFT}}^{125}$, but assumes light neutralinos and charginos similar to $M_h^{125}(\tilde{\chi})$ with respect to M_h^{125} .

The cross sections and branching ratios are provided for all of these MSSM scenarios as a function of m_A and $\tan \beta$. The Higgs masses and mixings, as well as the effective Yukawa couplings, are calculated with FEYNHIGGS [58–64]. The branching ratios are also determined from FEYNHIGGS, along with additional contributions from HDECAY [65, 66] and PROPHECY4F [67, 68].

The ggF cross sections are obtained with SUSHI, as for the THDM scenarios. For the MSSM scenarios other than $M_{h,\text{EFT}}^{125}$ and $M_{h,\text{EFT}}^{125}(\tilde{\chi})$, SUSHI includes NLO supersymmetric-QCD corrections [69–71] in expansions of heavy SUSY masses. The VBF cross sections are calculated in the same way as for the THDM scenarios. In the MSSM, α is determined as a function of m_Z , m_A and β [72].

5 Backgrounds

The following backgrounds are important for this analysis:

5.1 Top background

Background events involving at least one top quark are referred to as the top background. The largest contribution in the top background arises from $t\bar{t}$ events. A smaller contribution is due to single-top quark events, including the s-channel, t-channel and tW production modes.

As there are b quarks present in $t\bar{t}$ and single-top quark events, the identification of these b quarks using b -tagging algorithms is used in the analysis to reject a large fraction of the top background.

Top pair MC events are simulated at NLO and are reweighted to NNLO QCD + NLO electroweak (EWK) precision using the theoretical cross sections from Ref. [73]. This is done based on MC events simulated with the CP5 PYTHIA tune. For the 2016 data taking conditions, where a different tune is used, an additional weight is applied to take the differences between the tunes into account.

5.2 WW background

The non-resonant production of a pair of W bosons is another major source of background in the analysis. It is split into five different types:

- **qWW**: Two W bosons are produced through quark-initiated processes. Only QCD vertices are considered in higher orders. Events from this MC sample are simulated at NLO accuracy and are weighted by the generator level $p_{T,WW}$ to NNLO + next-to-next-to-leading logarithmic (NNLL) QCD accuracy [74, 75] to give a better agreement with data. This leads to a reweighting for $p_{T,WW} < 500$ GeV of up to 3%. In order to avoid overlap with two-jet events by the $qqWWqq$ and $WW2J$ background contributions described below, events from the qWW background that have at least two jets are removed if their generator level dijet mass is higher than 100 GeV.
- **ggWW**: The W bosons are produced through ggF . Events are simulated at LO, but the cross section of these events is reweighted to NLO by a k -factor of 1.53 [76]. The contribution from this process interferes with the ggF signal samples.
- **WWewk**: The W bosons are produced from vector boson scattering (VBS) at LO. Only EWK vertices are considered. Events containing top quarks or Higgs bosons at generator level are removed. The generator level mass of the two W bosons is required to be between 60 GeV and 100 GeV to avoid overlap with the DY and $V\gamma$ samples, where V is either a W or a Z boson.
- **qqWWqq**: The W bosons are produced at higher order from VBS, such that there are always at least two additional jets present. A cut on the generator level $m_{jj} > 100$ GeV is applied to avoid overlap from the qWW process. This type of WW background contains only events where both additional jets are quark jets. The contribution from this process interferes with the VBF signal samples.
- **WW2J**: This consists of events which are not included in the $qqWWqq$ background sample because they contain gluon jets. The requirement $m_{jj} > 100$ GeV is also applied.

5.3 DY background

The DY background is the dominant background in the SF final state. It is also relevant in the OF final state, where the Z boson decays into a pair of τ leptons, which further decay leptonically into an electron and a muon, along with the corresponding neutrinos.

In the OF final state, the DY background is determined through an embedding technique [77], where $Z \rightarrow \mu\mu$ events are selected from data. The muons are replaced by simulated tau leptons with the same four-momenta, which subsequently decay into an electron or muon. DY MC events are used if either final state lepton does not originate from a tau lepton, or if events were selected by a trigger other than the electron-muon trigger.

DY MC samples are used in the SF final state and have a number of corrections applied to them. One of these are the “recoil corrections”. The distribution of the parallel and perpendicular recoil components, defined as the missing transverse momentum p_T^{miss} components that are parallel and perpendicular to the momentum vector of the Z boson in the transverse plane, differs between data and MC. In DY events where only a single boson is produced in the event, this may have a strong effect on the measured p_T^{miss} . The analysis makes use of the corrections from Ref. [78].

Simulated DY events at both LO and NLO accuracy are used. We derived a reweighting to scale the distribution of the Z boson p_T ($p_{T,Z}$) in LO events to the corresponding one in NLO events. The weight was obtained using inclusive LO and NLO MC samples.

Another reweighting was derived to account for mismodeling of the $p_{T,Z}$ distribution at low p_T , which originates due to the effect of the underlying events and parton showers. Candidate $Z \rightarrow \mu\mu$ events are selected in data and MC in a phase space where the dilepton mass is near the Z boson mass. Only events with no additional jets with $p_T > 30$ GeV are considered. The reweighting functions are determined comparing the $p_{T,Z}$ spectrum in data and MC. This is done only up to $p_{T,Z} < 40$ GeV, as this region is known to be sensitive to higher order resummation effects that are not included in the simulation. At higher p_T the weight takes the same constant value as at 40 GeV. The weight is applied to the MC as a function of the generator level $p_{T,Z}$ in events with no generator level jets with $p_T > 30$ GeV.

The m_{ll} distribution of DY events is also corrected, as further disagreements with data were observed in the low m_{ll} region. Correction weights were determined using events with $p_T^{\text{miss}} < 30$ GeV, which mostly consist of DY events. The resulting reweighting functions were obtained as a function of m_{ll} between $20 < m_{ll} < 70$ GeV and are normalized such that the weight is 1.0 at $m_{ll} = 70$ GeV.

The DY MC samples also include events with an additionally emitted photon. Such events overlap with the $V\gamma/V\gamma^*$ MC samples (see below) and thus we remove events from the DY MC sample where at least one prompt photon is present at generator level with $p_T > 15$ GeV and $|\eta| < 2.6$ in addition to two prompt leptons of $p_T > 15$ GeV.

5.4 Non-prompt background

An additional source of background arises from the misidentification of leptons, primarily in W+Jets events. This background is estimated using a data-driven approach. A region enriched in QCD multijet events is selected from data, from which we extract a “fake rate” defined as the ratio of the number of leptons passing tight selection criteria over the number of leptons passing a looser selection. This is done using events where a jet above a given energy threshold is recoiling from a non-prompt lepton. Similarly, the prompt lepton efficiency is extracted using the tag-and-probe method by selecting events from data with a dilepton mass near the Z-pole. This efficiency and the fake rate are used to define a correlation between the total number of events with 2 loose leptons with either 0, 1 or 2 tight leptons, and the expected number of events with either 0, 1 or 2 non-prompt leptons.

5.5 Multiboson background

Background events with two or more bosons other than WW are here defined as the multiboson background. They include VZ, $V\gamma^*$ and VVV. All of these processes are simulated at NLO accuracy. The events from the VZ process are reweighted to NNLO using a k -factor of 1.11 [79].

In the $V\gamma^*$ background the photon (γ^*) is virtual and results in a pair of leptons. The $V\gamma^*$ MC

events are split into three categories. Two of the categories are based on matching dilepton pairs with the virtual photon (low mass, <4 GeV and high mass, >4 GeV). The third category consists of events where no matched dilepton pair is found ($V\gamma$ background events). From a measurement to data in a sideband region of events with three isolated leptons, we extracted k -factors of 0.94 for the low γ^* mass category and 1.14 for the high γ^* mass category [80].

5.6 SM Higgs background

The production of the SM Higgs boson is a background to this analysis. We consider contributions from $H \rightarrow WW$ and $H \rightarrow \tau\tau$ decays in the ggF, VBF and VH production modes, simulated at NLO accuracy.

6 Event selection and categorisation

The event selection requirements are designed to select signal events at a high efficiency whilst suppressing as many background events as possible. The selection criteria for all final states and categories are listed in Table 1. The various selections and categorizations implemented are now discussed in more detail.

6.1 Preselection

As part of the preselection, we require the two leptons passing tight identification requirements and sufficiently isolated from other particles within a cone of $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} = 0.3$ for electrons and $\Delta R = 0.4$ for muons. The leading lepton must have p_T above 30 GeV, while the subleading lepton must have p_T above 20 GeV. These requirements are significantly higher than the trigger thresholds of at most 23 GeV and 13 GeV for the leading and subleading leptons, respectively. Events containing a third lepton with $p_T > 10$ GeV and satisfying loose identification criteria are vetoed, this reduces background contributions from diboson events containing a Z boson or a photon, as well as any tri-boson events. As neutrinos are present in the final state of interest, we impose a requirement on the missing transverse momentum p_T^{miss} (MET), obtained from the “Pileup Per Particle Identification” algorithm [81]. The DY background is reduced by applying a cut on the transverse momentum of the dilepton system $p_{T, \text{ll}}$. A cut on the dileptonic mass m_{ll} is used to suppress the non-prompt background.

Additional requirements are applied to events in the SF final state to suppress the dominant DY background. The p_T^{miss} requirement is increased to 40 GeV while the requirement on the transverse mass of the dilepton and MET system is increased to $m_{T, \text{ll}+\text{MET}} > 80$ GeV. In addition, a cut is added on the transverse mass $m_{T, \text{ll}+\text{MET}}$ of the leading W boson reconstructed from only the leading lepton and the MET. Finally, events with dilepton invariant mass around the Z-peak ($70 \text{ GeV} < m_{\text{ll}} < 110 \text{ GeV}$) are removed from the signal region.

These are the baseline requirements used to search for an additional Higgs boson at low mass, close to the mass of the SM Higgs boson, up to about 1000 GeV. For higher signal masses we use an additional “high mass selection” which has been optimized for high mass resonances.

6.2 High mass selection

An additional set of cuts for the high mass selection is shown in Table 1. These are applied on top of the preselection requirements. The W bosons from the decay of a very high mass resonance are highly boosted. Therefore it is expected that the two final state charged leptons are detected on opposite sides of the detector. This motivates the requirement on the angle ϕ between these two leptons. A similar behavior is expected in the longitudinal plane along

Table 1: Summary of the selection criteria.

Preselection	1. Lepton $p_T > 30 \text{ GeV}$ 2. Lepton $p_T > 20 \text{ GeV}$ 3. Lepton $p_T < 10 \text{ GeV}$ (if present) Missing transverse momentum $p_T^{\text{miss}} > 30 \text{ GeV}$ Dilepton transverse momentum $p_{T,11} > 30 \text{ GeV}$ Dilepton mass $m_{\eta_1} > 20 \text{ GeV}$	
High mass selection	Angle ϕ between leptons $\phi_{11} > 2.0 \text{ rad}$ Deviation in ϕ and η between back-to-back leptons < 3.0 Angle ϕ between either lepton and MET $\phi_{1,\text{MET}} < 1.6 \text{ rad}$ Scalar sum of the transverse momenta $p_{T,11} + p_{T,12} + p_T^{\text{miss}} > 250 \text{ GeV}$ $m_{\eta_1} > 140 \text{ GeV}$	
Category	Opposite flavor	Same flavor
Signal region	Number of b-tagged jets $N_{b\text{-tag}} = 0$ Transverse mass $m_{T,11+\text{MET}} > 60 \text{ GeV}$	
		$m_{T,11+\text{MET}} > 80 \text{ GeV}$ $20 \text{ GeV} < m_{\eta_1} < 70 \text{ GeV}$ or $m_{\eta_1} > 110 \text{ GeV}$ Transverse mass of leading W boson $m_{T,11+\text{MET}} > 60 \text{ GeV}$ $p_T^{\text{miss}} > 40 \text{ GeV}$
Top control region	$N_{b\text{-tag}} \geq 1$	
	$m_{T,11+\text{MET}} > 60 \text{ GeV}$	$m_{T,11+\text{MET}} > 80 \text{ GeV}$ $20 \text{ GeV} < m_{\eta_1} < 70 \text{ GeV}$ or $m_{\eta_1} > 110 \text{ GeV}$ $m_{T,11+\text{MET}} > 60 \text{ GeV}, p_T^{\text{miss}} > 40 \text{ GeV}$
DY control region	$N_{b\text{-tag}} = 0$	
	$m_{T,11+\text{MET}} < 60 \text{ GeV}$ $m_{\eta_1} < 90 \text{ GeV}$	No cut on $m_{T,11+\text{MET}}$ $70 \text{ GeV} < m_{\eta_1} < 110 \text{ GeV}$ $m_{T,11+\text{MET}} > 60 \text{ GeV}, p_T^{\text{miss}} > 40 \text{ GeV}$

the z -axis. We consider a variable $\sqrt{(\pi - \phi_{ll})^2 + (\eta_{l1} + \eta_{l2})^2}$ which is 0 for two perfectly back-to-back leptons. A loose cut on this variable further enhances the sensitivity for high mass signals. The requirement on the dilepton mass m_{ll} is tightened to significantly reduce the DY background in both the OF and SF categories. Finally, a cut on the scalar sum of the transverse momenta of the two leptons and MET was also found to reduce background further while having no effect on the high mass signal yields.

6.3 Signal region selection

Background from DY events can be greatly reduced by adding a cut on the reconstructed transverse mass $m_{T,ll+MET}$ of the dilepton system and the MET. Together with the cuts on MET and on $p_{T,ll}$, it insures the QCD multijet background is suppressed to a negligible level. As there are no b quarks in the final state of interest, events containing at least one jet with $p_T > 20$ GeV and $|\eta| < 2.5$ passing the loose b -tagging working point of the Deep CSV discriminator [82] are vetoed. This significantly reduces the contribution of background events involving top quarks.

6.4 Top control region

The top control region is defined identically to the signal region, except for the requirement on the number of b -tagged jets. Only events containing b -tagged jets with $p_T > 30$ GeV are selected. This control region is used in the analysis to determine the normalization of the top background. The top control region is defined using the preselection requirements without the additional high mass selection. This is done to preserve the statistical power of the control region.

6.5 DY control region

The DY control region is selected identically as the signal region, except for the cut on the transverse mass $m_{T,ll+MET}$. To increase the purity of DY events in the OF final state, a cut on the dilepton mass $m_{ll} < 90$ GeV is added. For the SF final state, events with dilepton mass around the Z-Pole, $70 \text{ GeV} < m_{ll} < 110 \text{ GeV}$, are selected. This control region is used to constrain the normalisation of the DY background. As for the top control region, the preselection requirements without the additional high mass selection are used.

6.6 Sub categorization using a Deep Neural Network

Events in the signal regions and the control regions are further divided into the following three categories: “ggF”, “VBF” and “background”. The classification is performed by a Deep Neural Network (DNN). The ggF and VBF categories contain events that have a ggF-like or VBF-like signature. Events that are not classified as either signal are sorted into the background category.

For the training of the DNN, we used signal samples across the whole mass range. Only the most relevant backgrounds were used for the training. These include top, DY and non-resonant WW events. All events are weighted such that the effective number of events used in the training is the same for both production modes and across all available signal masses and backgrounds.

The output layer of the DNN has three nodes corresponding to the ggF signal, the VBF signal, and the background. The softmax activation function is used in the output layer. This normalizes the output such that the sum of all nodes is 1, corresponding to the probability of an event being a ggF or VBF signal, or a background process. The loss function used in this DNN is the “categorical crossentropy” function by Scikit-learn. The variables used as input are:

- The transverse and longitudinal components p_x, p_y, p_z of the momentum of the two leading leptons
- p_x, p_y, p_z and mass of the four leading jets, if present
- Dijet mass m_{jj} and pseudorapidity separation $\Delta\eta_{jj}$ for each combination of the four leading jets
- Total number of jets N_{jet}
- Number of jets with at least $p_T > 30 \text{ GeV}$
- p_x and p_y of the missing transverse momentum p_T^{miss}
- Reconstructed Higgs mass m_{reco}
- Dilepton mass m_{ll}
- Scalar sum H_T of the transverse momentum of all jets (with $p_T > 30 \text{ GeV}$) and leptons in the event, plus p_T^{miss}
- p_x and p_y of the vectorial sum H_T of the transverse momentum of all jets (with $p_T > 30 \text{ GeV}$) and leptons (with $p_T > 10 \text{ GeV}$) in the event
- Transverse masses $m_{T, ll+\text{MET}}, m_{T, l1+\text{MET}}$ and $m_{T, l2+\text{MET}}$ between the dilepton system and the MET, the leading lepton and the MET, and the subleading lepton and the MET, respectively
- Dilepton transverse momentum $p_{T, ll}$
- Angular correlations $\phi_{l1, \text{MET}}$ and $\phi_{l2, \text{MET}}$ between each of the two leptons and the MET
- Angular correlations ϕ_{ll} between the two leptons

For any jets that do not exist the value 0 is used to fill the corresponding p_x, p_y, p_z, m_{jj} and $\Delta\eta_{jj}$ variables in the event information. The variable m_{reco} listed above is constructed from the MET and the kinematics of the leptons, including their z -component:

$$m_{\text{reco}} = \sqrt{(E_{ll} + p_T^{\text{miss}})^2 - (\vec{p}_{ll} + \vec{p}_T^{\text{miss}})^2} \quad (1)$$

The primary goal of this DNN is to separate VBF signal events from ggF signal events. This is possible because the additional jets from VBF production have relatively large values of m_{jj} and $\Delta\eta_{jj}$. Furthermore, background events do not share this feature with VBF events. It is then possible to use the DNN output scores to effectively sort VBF events from ggF and background events. We then apply a further requirement to distinguish the remaining ggF and background events from each other. Since the ggF signal within a resonance mass range between 200 GeV and 300 GeV shares many similarities with the non-resonant WW background, a perfect separation is not possible. For this reason events sorted into the background category are not discarded.

7 Discriminating variable

For the final discriminating variable used in this analysis we use a DNN which attempts to reconstruct the resonance mass of a signal candidate. To obtain a mass hypothesis as the output of the DNN, the output layer has a single node and uses a linear activation function. The mean squared error is used for the loss function. This DNN uses the same input variables as shown above, except for the following variables which were found to have no significant impact on the separation of different signal mass hypotheses:

- p_x, p_y, p_z and mass of the third and fourth leading jet
- Dijet variables m_{jj} and $\Delta\eta_{jj}$ for each combination of the jets
- Total number of jets N_{jet}
- Number of jets with at least $p_T > 30 \text{ GeV}$
- Dilepton transverse momentum $p_{T,\ell\ell}$
- Angular correlations $\phi_{\ell\ell}$ between the two leptons

It was found that the discriminating variable obtained from the DNN, “DNN m_T ”, is closer to the true resonance mass of the signal than the m_{reco} variable used in previous high mass analyses in this channel. The final signal extraction fit to data is therefore performed on this variable, allowing for good signal to background discrimination. Figures 1, 2 and 3 shows the DNN m_T distributions observed, along with the full background estimation and the predicted contribution of a 1000 GeV signal. The uncertainty band on the background estimation corresponds to the total systematic uncertainty on the templates after the fit to data.

8 Uncertainties

An overview of all uncertainties is given in Table 2, and a full description of each given below.

- **Floating normalization:** For the top and DY backgrounds, as well as parts of the WW background (qWW and WW2J), the normalization of the estimated background is left floating during the final fit to the data. The top and DY control regions are used to constrain these normalizations. The WW background normalization is primarily constrained by the background categories.

All normalizations are uncorrelated between all data-taking years, final states and for each category, except for the WW background, since there is no dedicated WW control region.

- **Luminosity:** A luminosity uncertainty is applied on the overall normalization of all signals and backgrounds, consisting of an uncorrelated and a correlated year by year component [83–85]. Overall the luminosity uncertainties are 1.2% in 2016, 2.3% in 2017, and 2.5% in 2018.
- **Trigger efficiency:** The trigger efficiency is calculated from the tag-and-probe method. The systematic uncertainty is determined by varying the p_T requirement on the tagged leptons by $\pm 5 \text{ GeV}$, and by varying the Z mass window range from 60–120 GeV to 70–110 GeV. The resulting uncertainty is added in quadrature with the statistical uncertainty. This leads to an overall uncertainty of about 0.2%.
- **Lepton efficiency/scale:** The lepton efficiency uncertainty is about 1.3% for electrons in 2016 and 2018, and about 1.7% for electrons in 2017. For muons it is about 1.0%, 1.1% and 0.9% in 2016, 2017 and 2018, respectively. For the momentum scale an uncertainty of 0.2% is considered for electrons with $p_T < 500 \text{ GeV}$, while 0.5% is taken for high p_T electrons. The muon momentum scale uncertainty is 0.2% [86].
- **Jet energy scale:** Individual components [18] of the jet energy scale uncertainty are considered, with none of them surpassing an uncertainty of 2% for events in the signal region.
- **Jet energy resolution:** Corrections on the resolution of the jets are applied in 2017 and 2018. The corresponding uncertainties vary the p_T of the jets by 12.5% on average in 2017 and 13.0% in 2018.

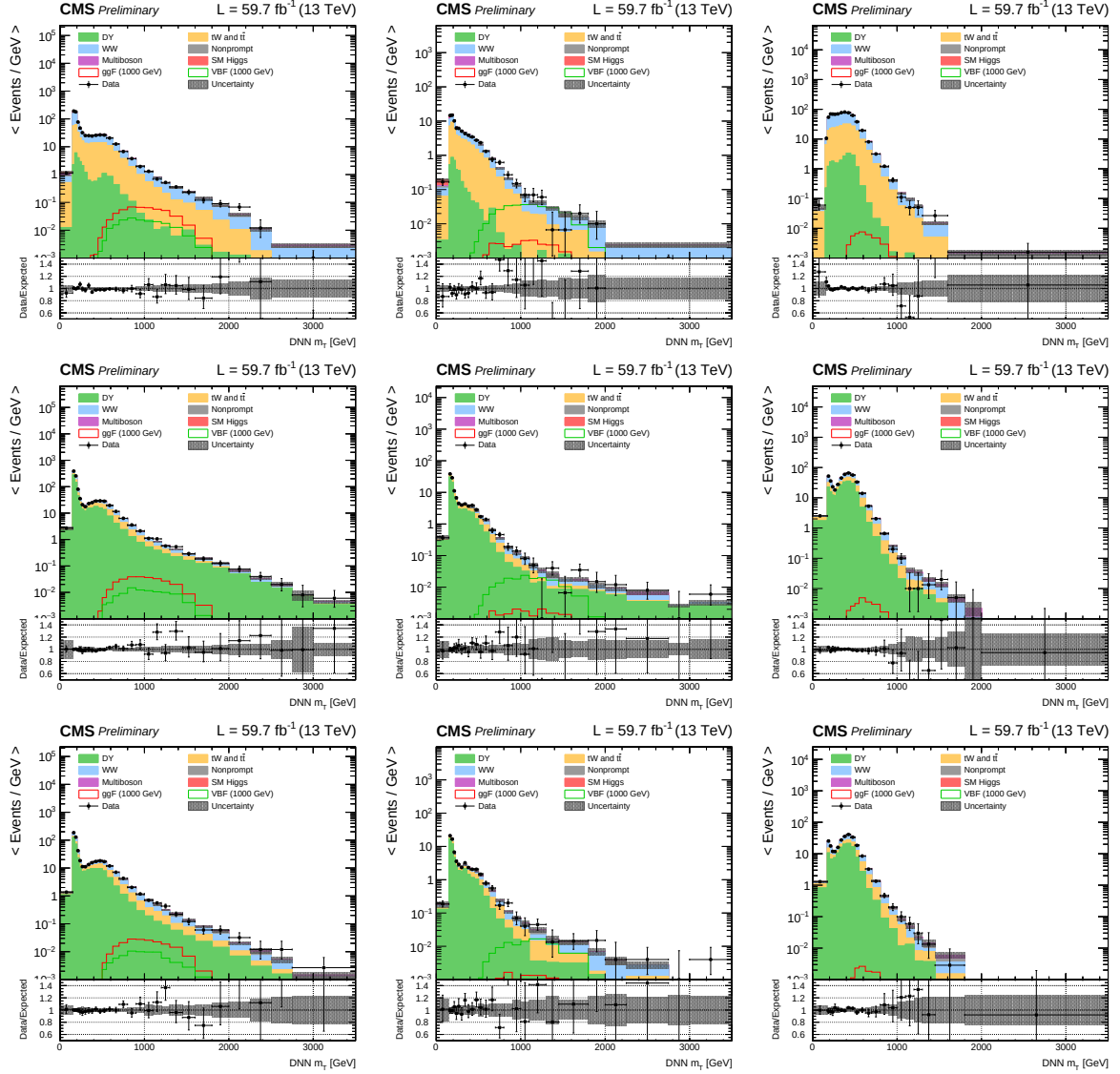


Figure 1: DNN m_T distributions for the 2018 data set along with the background estimation and the prediction of a 1000 GeV signal, for events passing the e μ (top), $\mu\mu$ (middle) and ee (bottom) signal region selections and entering the ggF (left), VBF (middle) and background (right) categories.

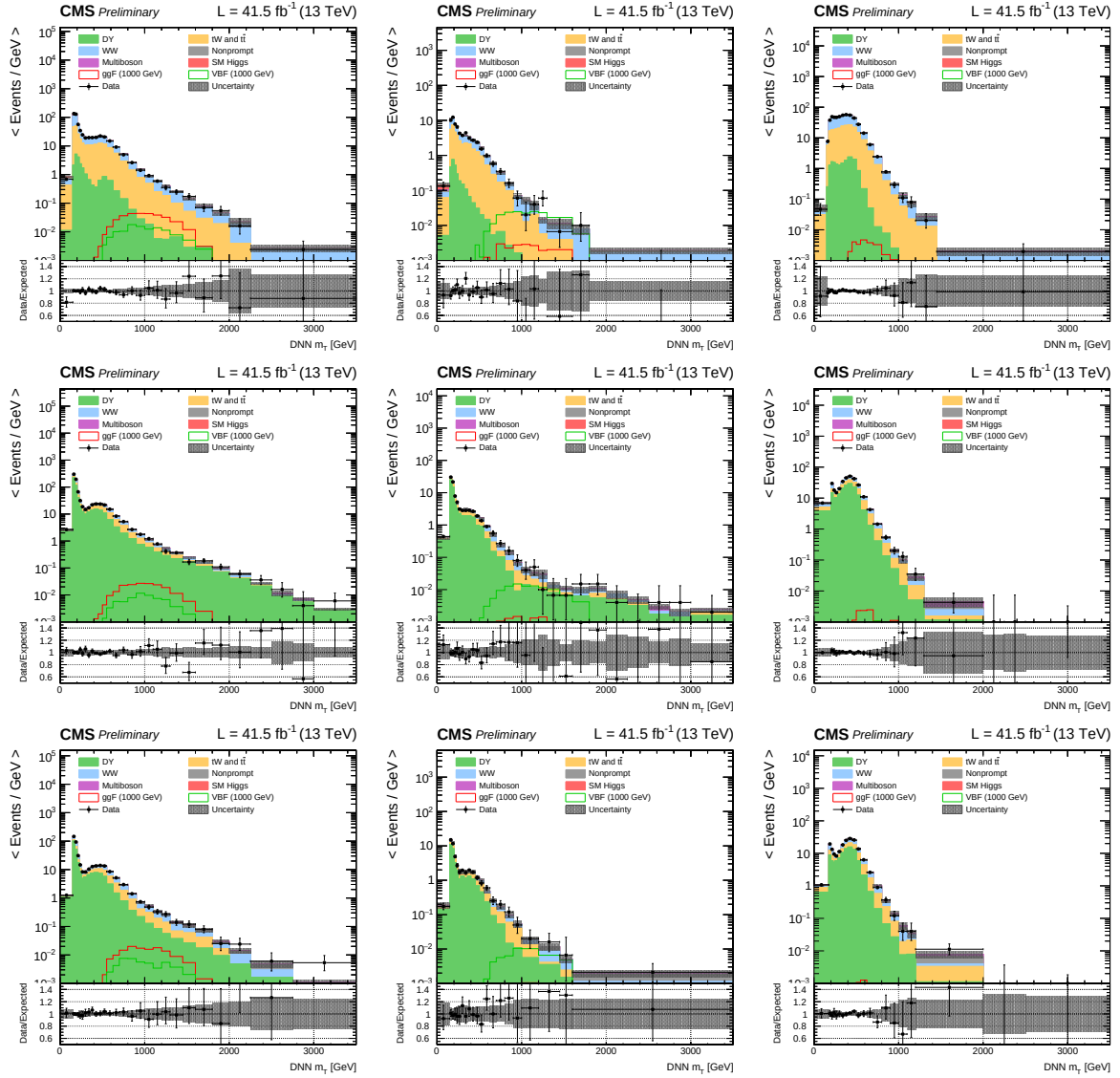


Figure 2: DNN m_T distributions for the 2017 data set along with the background estimation and the prediction of a 1000 GeV signal, for events passing the $e\mu$ (top), $\mu\mu$ (middle) and ee (bottom) signal region selections and entering the ggF (left), VBF (middle) and background (right) categories.

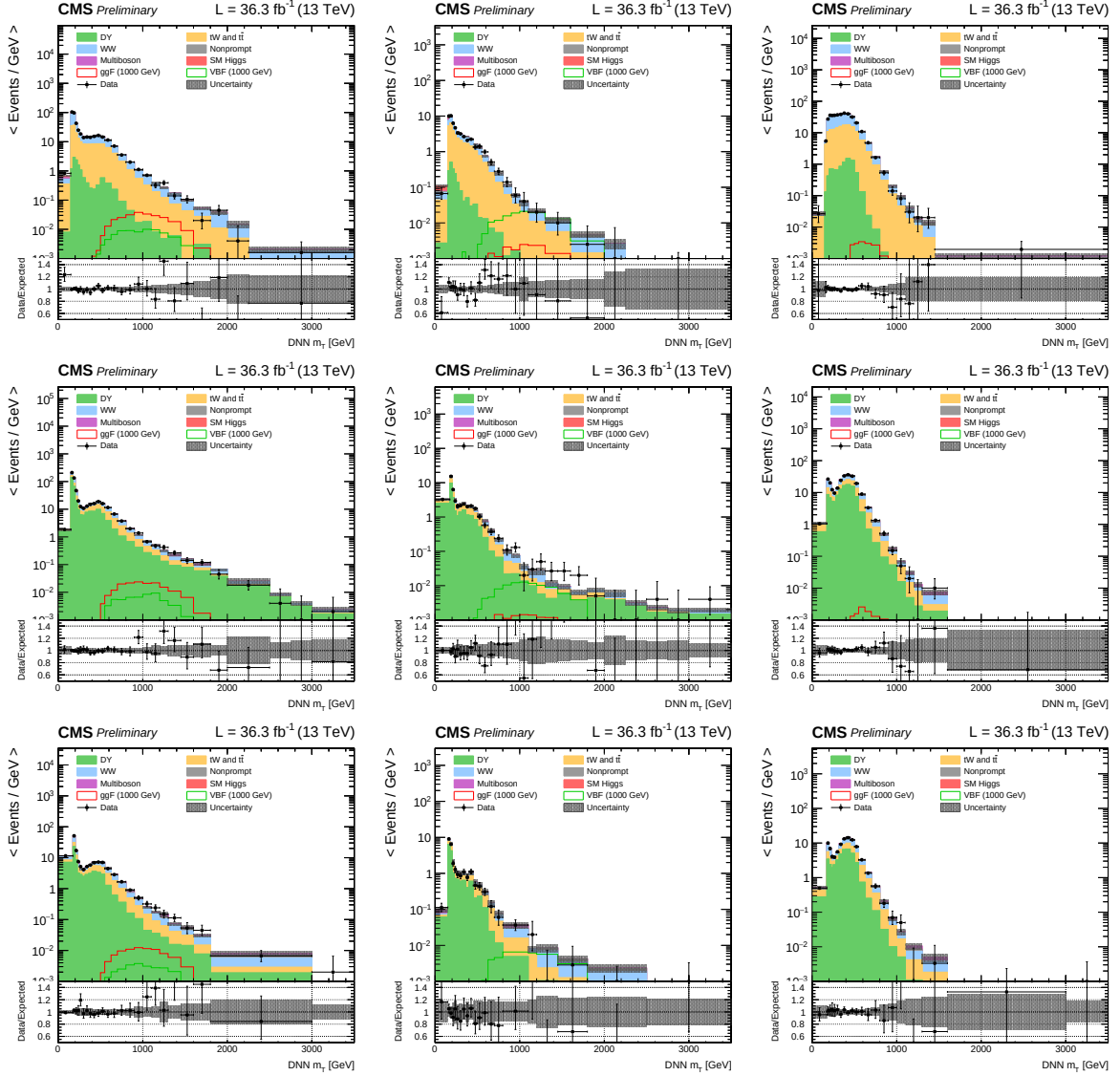


Figure 3: DNN m_T distributions for the 2016 data set along with the background estimation and the prediction of a 1000 GeV signal, for events passing the $e\mu$ (top), $\mu\mu$ (middle) and ee (bottom) signal region selections and entering the ggF (left), VBF (middle) and background (right) categories.

- **b-tagging:** Uncertainties on the b-tagging scale factors come from a number of different sources. These include contributions due to the jet energy scale uncertainty, cross contaminations from light and heavy flavored jets and statistical fluctuations.
- **MET energy scale:** The MET scale uncertainty is determined by adding or removing the unclustered energy, and by propagating the electron/muon/jet energy scale uncertainties to MET. Additional uncertainties on the MET response and resolution are considered for the recoil corrected DY MC samples.
- **Prefire correction:** Corrections that fix the disagreement between data and MC due to “prefiring” of the trigger are added to the MC samples for 2016 and 2017 conditions. The corresponding uncertainty is on average 0.4% for 2016 and 0.8% for 2017.
- **Parton shower (PS):** The PS uncertainties are split into initial state radiation (ISR) and final state radiation (FSR) components. They are determined by PYTHIA 8 in the form of event weights and correspond to variations of the ISR and FSR scales by a factor of 0.5 or 2.
- **Underlying event (UE):** The systematic uncertainty due to UE is derived from different PYTHIA 8 tunes and was found to be 1.5%.
- **Pileup (PU):** The minimum bias cross section of 69.2 mb^{-1} is varied by 4.6%.
- **Pileup jet ID:** An uncertainty is added on the additional identification criterion used to reduce the contamination of jets from PU [87].

Uncertainties on the non-prompt lepton background estimation :

- **Systematical:** The data set used to determine the fake rates may have different sources (different flavored jets) of non-prompt leptons than those in the signal region. An overall uncertainty of 30% is applied to the normalization of events with non-prompt leptons to take this effect into account.
- **Statistical:** The statistical uncertainty in the determination of the fakes rates is 10% and 4% for electrons and muons, respectively.
- **Jet energy thresholds:** The energy threshold of the jet recoiling from the fake lepton is varied by $\pm 10 \text{ GeV}$ to account for any differences between the reconstructed p_T of the fake lepton and its original jet (which should be within the range of the recoiling jet). This leads to an uncertainty of up to 1.6% and 5.2% for electrons and muons, respectively.

Uncertainties on the τ -embedded DY estimation in the OF final state:

- **Trigger, Electron and Muon efficiency/scale:** These are the same as for MC samples.
- **Double counting:** The τ -embedded events originally selected from data are not exclusively from true $Z \rightarrow \mu\mu$ events. There is also a small contamination from $t\bar{t}$ and diboson events. Double counting between these additional events and those from the corresponding MC samples is handled by vetoing MC events where the final state leptons originate from τ leptons. The uncertainty of this procedure is estimated as 10% of the distribution of the vetoed MC events.

Theoretical uncertainties:

- **QCD scale:** For ggF and VBF Higgs boson samples the scale uncertainties are taken from the LHC Higgs Cross Section Working Group [44]. For background samples that involve final state particles coming from either one or two bosons the scale un-

certainties are determined from varying combinations of different renormalization and factorization scales. The uncertainty is taken from the combination yielding the largest variation of the overall normalization. For the ggWW and WWewk samples we apply an uncertainty of 15% and 11%, respectively.

- **PDF:** The PDF uncertainties are also taken from the LHC Higgs Cross Section Working Group for all Higgs boson samples. For the most important background processes, the top, DY and qWW backgrounds, the PDF uncertainties are determined from MC samples with varied PDF sets. For other diboson backgrounds the uncertainties are within the range 4–5%.
- **QCD scale uncertainty on the acceptance:** QCD scale uncertainties were found to have an impact of 0.3–1.5% on the acceptance for all processes.
- **PDF uncertainty on the acceptance:** The effect on the acceptance of using different PDFs is 0.1–0.6% for all processes.

Other uncertainties:

- **Signal/Control region ratio:** We assign an additional uncertainty to the DY and top backgrounds of 2% and 1% within their respective control regions.
- **DY/top corrections:** An uncertainty is added for each of the corrections applied to the DY samples in the SF final states, as well as for the $t\bar{t}$ reweighting. The uncertainties for the p_{TZ} , NLO and m_{ll} DY reweighting are up to 14%, 26% and 35%, respectively, while for the $t\bar{t}$ reweighting the uncertainty is up to 16%.
- **WW resummation/scale variations:** The QCD resummation uncertainty is up to 5.7% in both the ggF and VBF categories. The QCD scale uncertainty is less than 1% in the ggF category and up to 5.2% in the VBF category. Uncertainties are correlated between years, but uncorrelated between categories.
- **$V\gamma^*$ normalization:** The MC $V\gamma^*$ contribution in the low γ^* mass region has an uncertainty of 25% from the comparison with data. In the high γ^* mass region the uncertainty is 16%.
- **Single-top quark:** We assign an uncertainty of 8% to account for possible differences between the $t\bar{t}$ and single-top quark contributions.
- **WW EWK NLO contribution:** An uncertainty is included to account for the contribution of NLO EWK effects in the qWW background sample. The size of this uncertainty is about 1%.
- **Statistical uncertainties:** The statistical uncertainty due to the limited number of simulated events is associated to each bin of the signal and background predictions.

9 Extrapolation analysis

A study is performed to estimate the expected sensitivity of this analysis at the high luminosity LHC (HL-LHC). For this purpose the signal and background expectations for Run 2, corresponding to an integrated luminosity of 138 fb^{-1} , are extrapolated to correspond to the expected integrated luminosity at the HL-LHC of 3000 fb^{-1} . The corresponding statistical uncertainties are scaled by $1/\sqrt{3000/138}$.

For the treatment of the systematic uncertainties, three scenarios are considered:

- **“Run 2”:** All systematic uncertainties from the Run 2 analysis are kept and extrapo-

Table 2: Considered systematic uncertainties and their year-by-year correlations.

Uncertainty	Process	Type	Correlation
Floating normalization	top, qWW, WW2J, DY	norm.	Uncorrelated; qWW/WW2J correlated between final states
Luminosity	All MC	norm.	Partially correlated
Trigger efficiency	All MC and Embedded	shape	Uncorrelated
Electron / Muon ID efficiency	All MC and Embedded	shape	Uncorrelated
Electron / Muon p_T scale	All MC and Embedded	norm.	Uncorrelated
Jet energy scale / resolution	All MC	norm.	Partially correlated
MET energy scale	All MC	norm.	Uncorrelated
Prefire corrections	All MC	shape	Uncorrelated
Parton shower	All MC	shape	Correlated between years, split between ISR and FSR
Underlying event	All MC	norm.	Correlated between 2017 and 2018
Pileup	Signal, top, DY, qWW, SM ggHWW, SM VBFWW	shape	Uncorrelated
Pileup jet ID	All MC	shape	Uncorrelated
Fake rate systematic	Fakes	norm.	Correlated between years, split between final states
Fake rate statistic	Fakes	shape	Uncorrelated
Fake rate jet threshold	Fakes	shape	Uncorrelated
Embedded double counted events	Embedded	shape	Uncorrelated
B-tagging	All MC	shape	Partially correlated
QCD scale variation	Signal, SM	norm.	Correlated between years, split between production modes
PDF variation	All other MC except qWW and VVV	shape	
QCD scale / PDF acceptance	All MC except VVV	norm.	Correlated between years, split between production modes
Control region acceptance	Signal, SM, ggWW, qqWWqq	norm.	Correlated between years, split between production modes
DY reweightings	top, DY	norm.	Correlated
$t\bar{t}$ reweighting	DY	shape	Uncorrelated
WW resummation / scale	top	shape	Correlated
$V\gamma^*$ normalization	qWW	shape	Correlated between years, split between categories
Single-top quark contribution	$V\gamma^*$	norm.	Correlated
EWK NLO correction	top	shape	Correlated
	qWW	shape	Correlated

lated to 3000 fb^{-1} along with the process yields. This scenario probes the increase in sensitivity from the improvements in the statistical precision alone.

- **“YR18”**: Compared to the “Run 2” uncertainty scenario, some systematic uncertainties are either downscaled or fully dropped. This scenario is a more realistic representation of the HL-LHC conditions.
- **“Stat. only”**: All systematic uncertainties are removed, thus only statistical uncertainties are included. This scenario represents a best-case scenario.

The changes of the systematic uncertainties under the YR18 scenario are given below:

- **Luminosity**: The uncertainties on the luminosity are replaced by a single uncertainty of 1%, correlated between all years.
- **Lepton efficiency**: The electron and muon efficiency is reduced by 50% for each lepton.
- **Jet energy scale**: The uncertainty on the absolute scale is reduced by 80%. Uncertainties on the jet flavor and those that are split between different parts of the detector are reduced by 50%. The uncertainty on relative sample efficiencies is removed.
- **MET energy scale**: The scale uncertainty on MET is reduced by 90%.
- **Prefire correction**: The uncertainty on prefiring is reduced by 80%.
- **Pileup**: The PU uncertainty is removed.
- **Pileup jet ID**: The uncertainty on the Jet PU ID is reduced by 50%.
- **$t\bar{t}$ reweighting**: The uncertainty on the $t\bar{t}$ reweighting is reduced by a third.
- **Statistical**: Any statistical parts of systematic uncertainties, e.g. non-prompt lepton background estimation and parts of the b-tagging uncertainties, are removed.
- **Theoretical**: The theoretical PDF, QCD scale uncertainties and the WW resummation and scale variations are reduced by 50%.

10 Results

The statistical methodology used to interpret the data was developed by the ATLAS and CMS collaborations in the context of the LHC Higgs Combination Group [88]. For each signal mass hypothesis a maximum likelihood fit of the data in all channels and categories is performed using a background-only and a signal-plus-background model. A small excess of data over the background expectation between DNN m_T values of 400 and 1200 GeV is observed. For each of the f_{VBF} scenarios considered, the signal hypothesis with the highest local significance is determined. The results of this study are given in Table 3. The highest local significance, 3.80σ , is found for the $f_{VBF} = 1$ scenario and corresponds to a signal hypothesis with a mass of 650 GeV and a production cross section of about 0.16 pb. The global significance is 2.60σ and is calculated with the method of Ref. [89].

The following results are expressed as 95% confidence level (CL) upper limits on the product of the production cross section and branching ratio over the signal mass range. In addition, we set model dependent exclusion limits on the THDM and MSSM.

10.1 Cross section limits

Figure 4 shows the exclusion limits for each of the f_{VBF} scenarios. For these limits the high mass selection is used for signal masses above 1000 GeV in the SM f_{VBF} and $f_{VBF} = 1$ scenarios,

while for the other two scenarios they are used for signal masses of ≥ 1900 GeV. The red line in Figure 4 corresponds to the product of the cross section and branching ratio expected for a SM-like resonance.

The highest mass which is observed (expected) to be excluded at 95% CL for a heavy SM-like Higgs boson for each of the f_{VBF} scenarios is:

- SM f_{VBF} : 2100 (1800) GeV
- $f_{VBF} = 1$: 2100 (1900) GeV
- $f_{VBF} = 0$: 800 (1000) GeV
- floating f_{VBF} : 900 (1200) GeV

The highest mass exclusion is observed for the SM f_{VBF} and $f_{VBF} = 1$ scenarios, where the presence of a heavy SM-like Higgs boson is excluded up to 2100 GeV. A deviation between the observed and expected upper limits of just over 2σ can be seen between 500 GeV and 900 GeV for the $f_{VBF} = 1$ and the SM f_{VBF} scenarios. In the other two scenarios a similar but smaller deviation can also be seen in the same mass region. They correspond to the data excess discussed in the previous section.

The 95% CL exclusion limits on the product of the cross section and branching ratio are also calculated for each f_{VBF} scenario as a function of the relative width of the signal. These are shown in Figure 5.

Table 3: Summary of the signal hypotheses with highest local significance for each f_{VBF} scenario. For each signal hypothesis the resonance mass, production cross sections, and the local and global significances are given.

Scenario	Mass [GeV]	ggF cross sec. [pb]	VBF cross sec. [pb]	Local signi. [σ]	Global signi. [σ]
SM f_{VBF}	800	0.16	0.057	3.2	1.7 ± 0.2
$f_{VBF} = 1$	650	0.0	0.16	3.8	2.6 ± 0.2
$f_{VBF} = 0$	950	0.19	0.0	2.6	0.4 ± 0.6
floating f_{VBF}	650	2.9×10^{-6}	0.16	3.8	2.4 ± 0.2

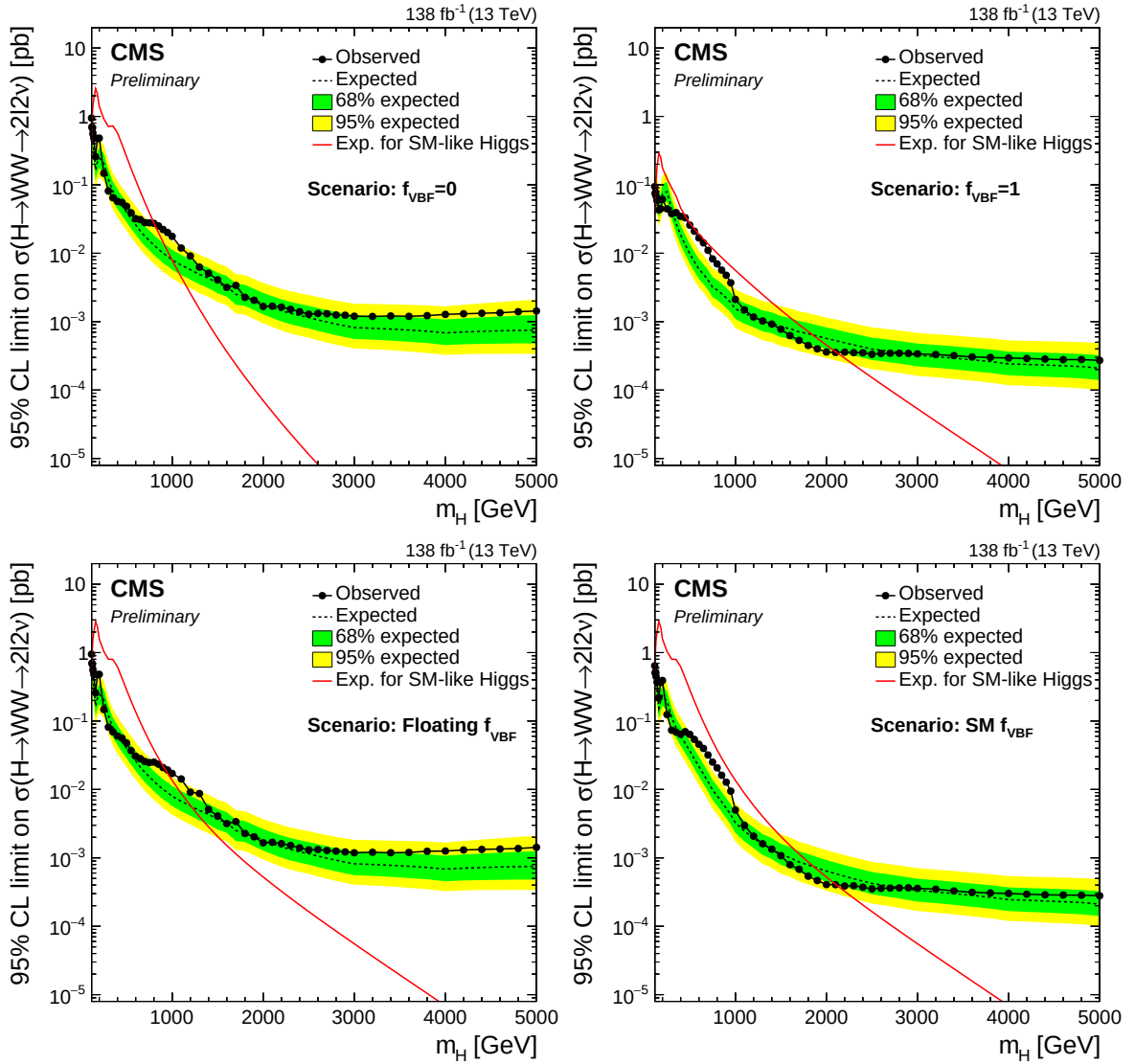


Figure 4: Limits using the combined Run 2 data set for the $f_{VBF} = 0$ (top left), $f_{VBF} = 1$ (top right), floating f_{VBF} (bottom left) and the SM f_{VBF} scenarios (bottom right).

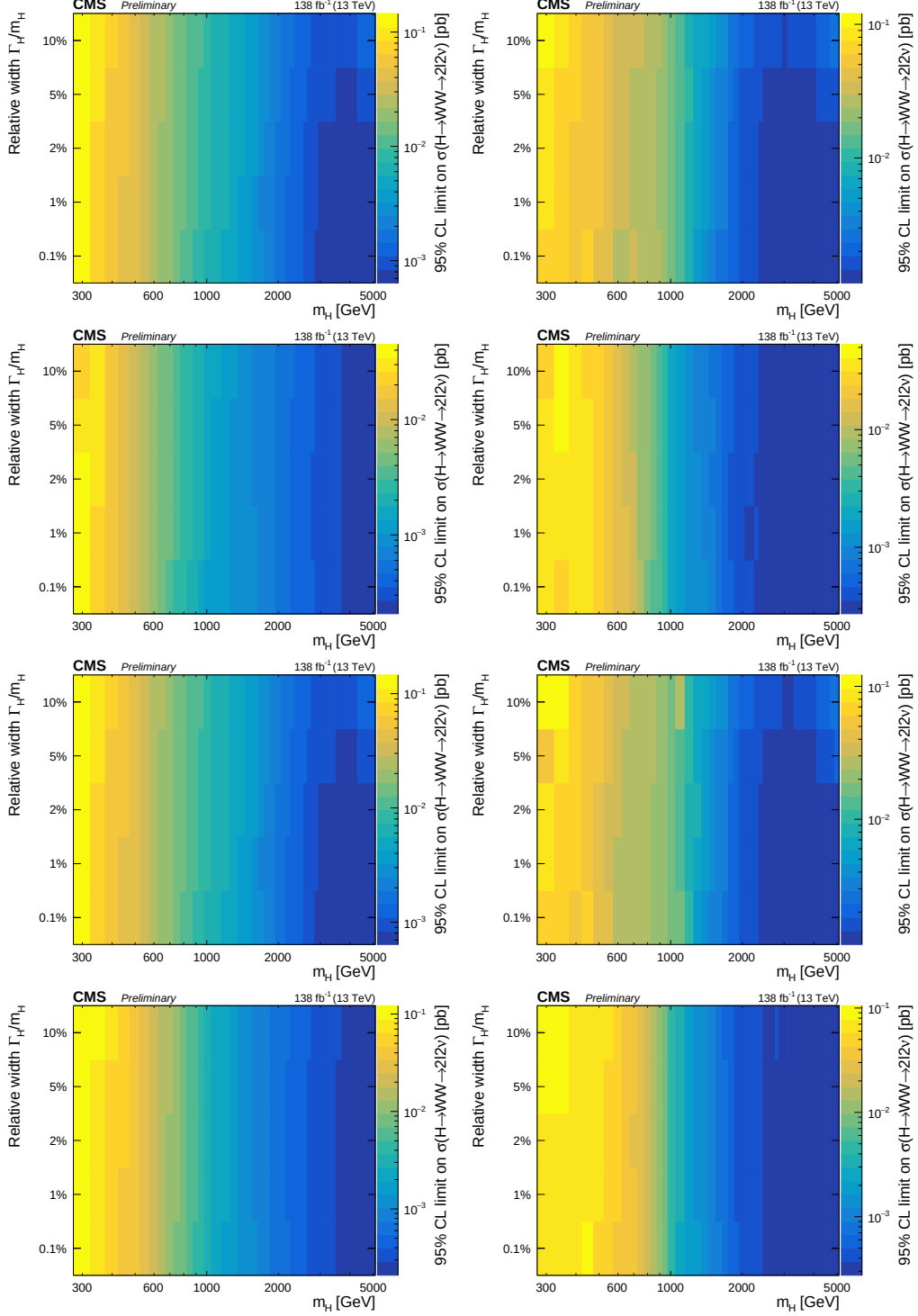


Figure 5: Expected (left) and observed (right) upper limits as a function of the signal mass and the relative width. From top to bottom, the limits are shown for the $f_{VBF} = 0$, $f_{VBF} = 1$, floating f_{VBF} and the SM f_{VBF} scenarios.

10.2 MSSM limits

The exclusion regions for the MSSM scenarios are shown in Figure 6. In general, the analysis is sensitive in the region at low m_A and $\tan\beta$. As m_A becomes larger the decoupling limit is approached, the two CP-even Higgs bosons h and H decouple and only h couples to a pair of vector bosons. For larger values for $\tan\beta$ the branching ratios of H to down-type quarks and leptons become dominant, thus the sensitivity in the WW decay mode decreases. Another feature of the curves is the “peak”-like structure seen around $m_A = 180$ GeV. At this point the corresponding mass of H is above 160 GeV. As a result, the sensitivity is increased as both W bosons are produced on-shell.

The observed exclusion regions are overall in agreement with the expected ones. At low $\tan\beta$ values, the highest excluded mass of A is in the range 400–450 GeV. At higher values of $\tan\beta$, the observed exclusion regions are slightly smaller than expected. At $\tan\beta = 10$, the highest excluded mass of A is in the range 100–150 GeV.

10.3 THDM limits

Figure 7 shows exclusion limits for a Type-II and Type-I THDM in the m_H - $\tan\beta$ -plane assuming $m_h = 125$ GeV, $m_H = m_A = m_{H^\pm}$ and $\cos(\beta - \alpha) = 0.1$. As in the MSSM scenarios, this analysis excludes the region at low values of $\tan\beta$. At $\tan\beta = 1$, it is expected that both scenarios are excluded up to a signal mass of about 900 GeV, but the observation shows only an exclusion up to about 700 GeV. The observed exclusion limits are however within the 2σ of the expectations. The exclusion limit for the Type-I THDM is similar to that of the Type-II THDM, except in the low mass region where the exclusion extends to larger values of $\tan\beta$.

Figure 8 shows the exclusion limits in the $\cos(\beta - \alpha)$ - $\tan\beta$ -plane for a number of signal mass hypotheses assuming $m_h = 125$ GeV and $m_H = m_A = m_{H^\pm}$. There is no exclusion at $\cos(\beta - \alpha) = 0$, as the coupling of H to two vector bosons vanishes.

10.4 Extrapolation results

The 95% CL expected exclusion limits, for the Phase-2 operations of the HL-LHC, on the product of the cross section and branching ratio for the SM f_{VBF} scenario are shown in Figure 9. Compared to the Run 2 limits, shown in Figure 4, the upper limits are improved by almost one order of magnitude. Except for low mass signals, the limits are insensitive to different uncertainty scenarios. This is because already for the Run 2 analysis the statistical uncertainties are dominant for higher mass resonances. The extrapolated limits for the other f_{VBF} scenarios can be found in the supplementary material.

The exclusions limits for the MSSM M_h^{125} scenario and the Type-II THDM are shown in Figure 10. The sensitivity in the MSSM scenario is only increased slightly, because the analysis quickly becomes less sensitive at higher values of m_A and $\tan\beta$ for the reasons discussed previously. In the Type-II THDM, we find a larger increase of the expected exclusion region. In addition to the exclusion up to almost $m_H = 1500$ GeV at low $\tan\beta$, the analysis starts to become sensitive to higher values of m_H above $\tan\beta = 10$. Although the production cross section of H is low in this region, the branching ratio of H into WW starts to increase. In all of these limits, it can be seen that the difference between the various uncertainty scenarios is more pronounced than for the model independent limits. The exclusion limits for the M_h^{125} (alignment) and $M_{h,EFT}^{125}$ scenarios are shown in the supplementary material.

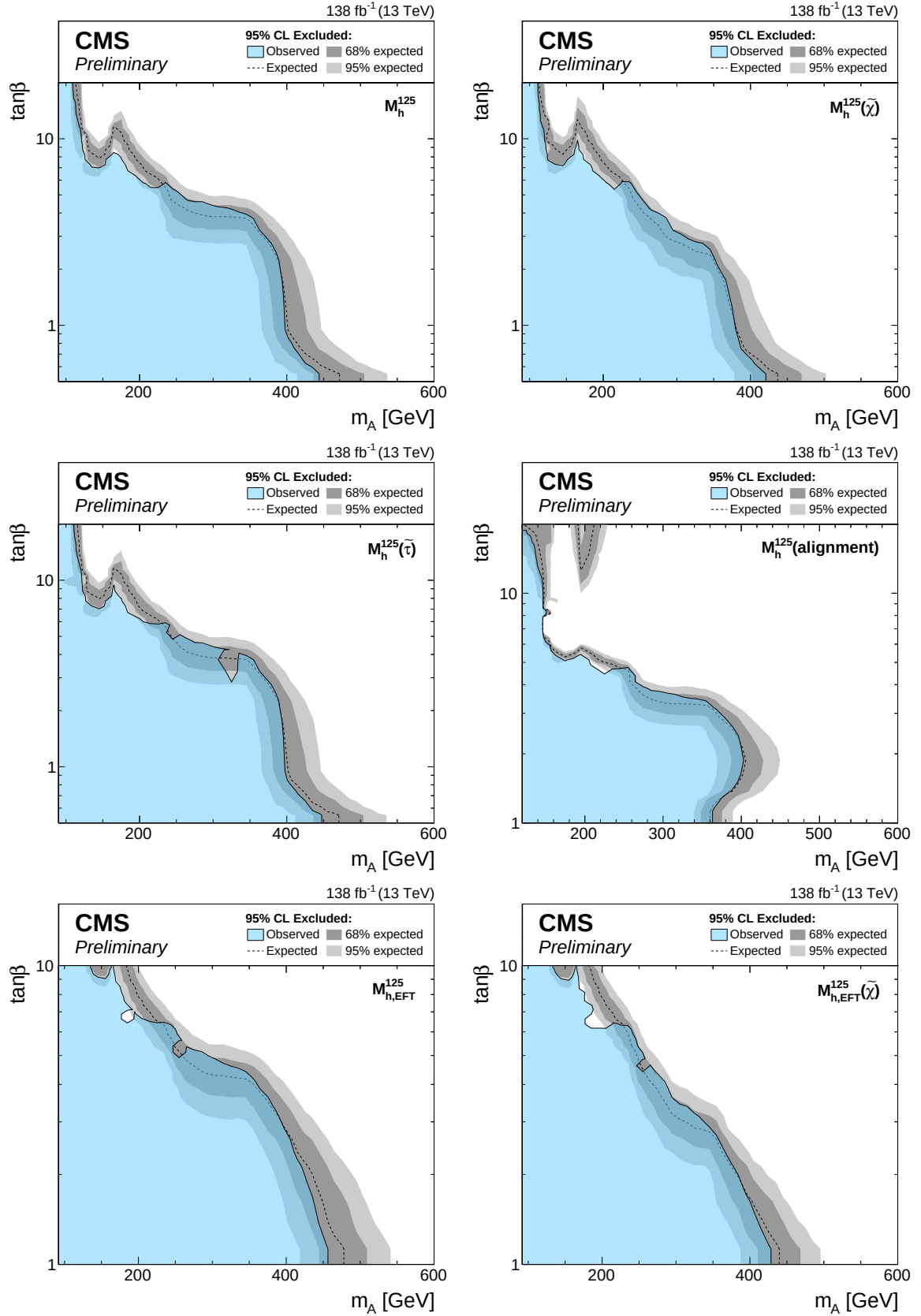


Figure 6: Expected exclusion limits for the MSSM scenarios. The limits are shown for the M_h^{125} (top left), $M_h^{125}(\tilde{\chi})$ (top right), $M_h^{125}(\tilde{\tau})$ (middle left), $M_h^{125}(\text{alignment})$ (middle right), $M_{h,\text{EFT}}^{125}$ (bottom left) and $M_{h,\text{EFT}}^{125}(\tilde{\chi})$ (bottom right) scenarios.

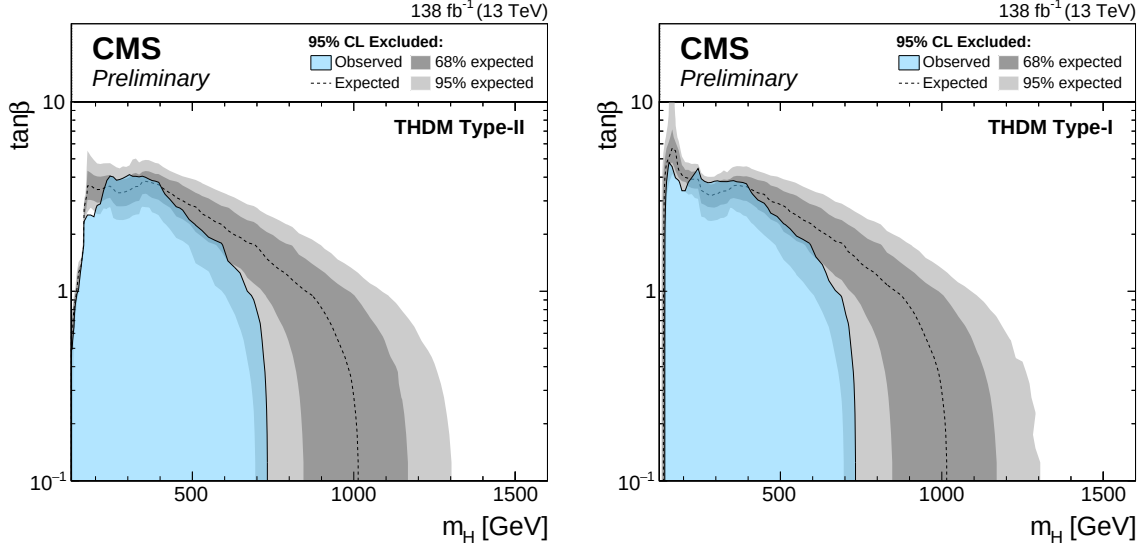


Figure 7: Exclusion limits for a Type-II THDM (left) and a Type-I THDM (right) over the m_H - $\tan\beta$ -plane for $\cos(\beta - \alpha) = 0.1$.

11 Summary

We performed a search for a high mass Higgs boson decaying into a pair of W bosons in the dileptonic channel. We observe an upward fluctuation of data compared to the expected background. The signal hypothesis with the highest significance corresponds to a resonance mass of 650 GeV in the scenario where only VBF production is considered. The global significance of this excess is 2.6σ .

The presence of a heavy SM-like Higgs boson is excluded at 95% CL up to 2100 GeV, assuming the relative contribution of ggF and VBF production is SM-like and also assuming only VBF production. The exclusion is up to about 800 GeV when considering only ggF production. In the case where the ratio between ggF and VBF production is left floating in the fit, an exclusion of up to 900 GeV is observed.

In MSSM scenarios the analysis is sensitive at low values of m_A and $\tan\beta$ and the exclusion limits extend up to $m_A = 450$ GeV. For m_A between 150 GeV and 400 GeV, we exclude values of $\tan\beta$ between 10 and 3. The sensitivity is similar between the M_h^{125} , $M_h^{125}(\tilde{\chi})$, $M_h^{125}(\tilde{\tau})$, $M_{h,EFT}^{125}$ and $M_{h,EFT}^{125}(\tilde{\chi})$ scenarios. In the M_h^{125} (alignment) scenario, the exclusion limits reach to about $m_A = 400$ GeV.

In THDM scenarios, the limits show an exclusion of up to 750 GeV in THDMs of both Type-I and Type-II. For $\cos(\beta - \alpha) = 0.1$, the limits in $\tan\beta$ extend up to 4 in Type-II and up to 5 in Type-I, with the sensitivity generally becoming lower for higher masses of H. The sensitivity over $\tan\beta$ varies more strongly as a function of $\cos(\beta - \alpha)$.

Extrapolation studies were performed to evaluate the expected gain in sensitivity for the Phase-2 operations of the HL-LHC. The upper limits on the product of the cross section and branching ratio of a new resonance are expected to be improved by almost one order of magnitude. The expected exclusion range in MSSM scenarios increases by only about 50 GeV along the m_A axis. The increase in sensitivity is more noticeable for a general THDM, with the exclusion extending from 1000 GeV up to 1500 GeV for low values of $\tan\beta$.

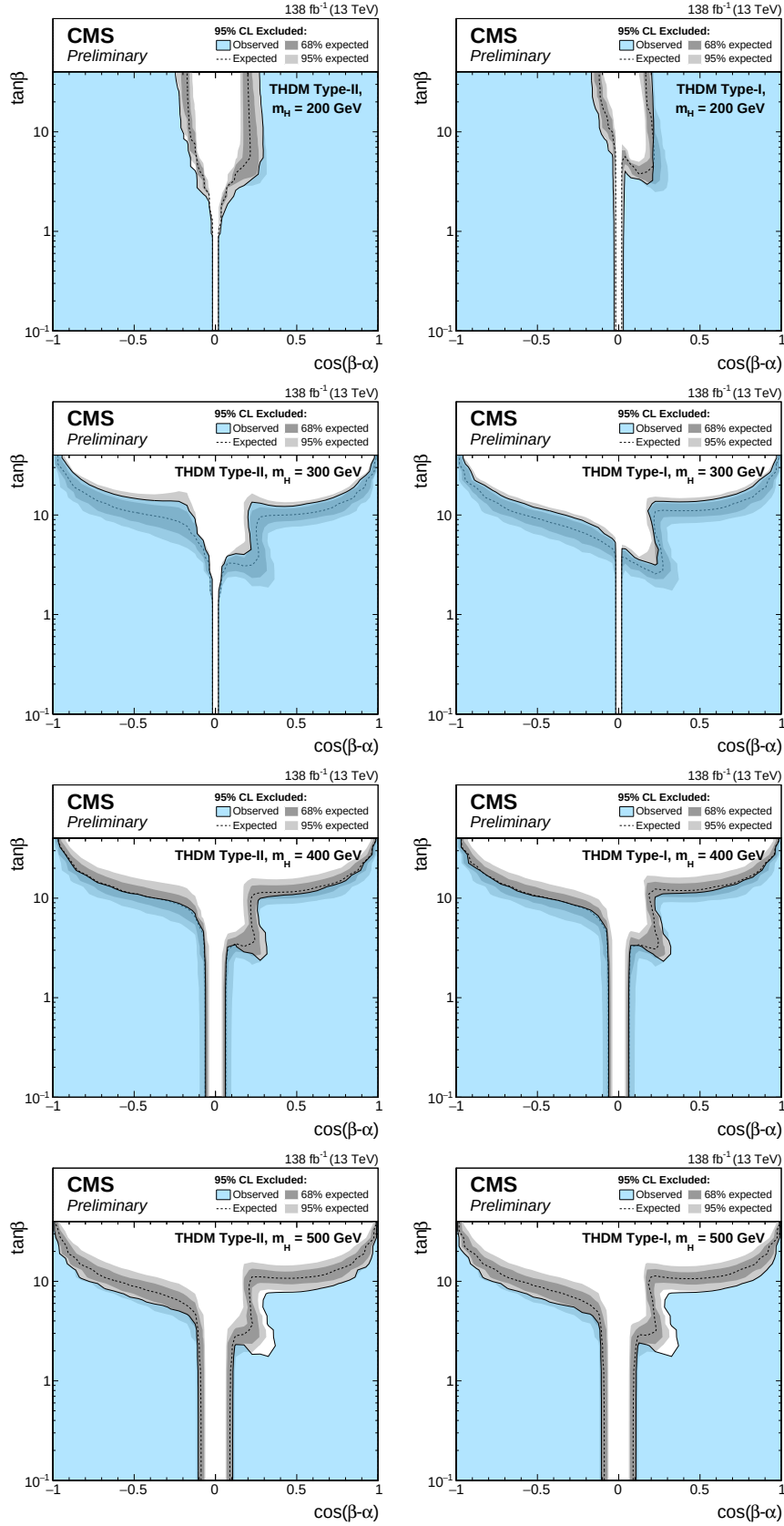


Figure 8: Exclusion limits for a Type-II THDM (left) and a Type-I THDM (right) over the $\cos(\beta - \alpha)$ - $\tan \beta$ -plane. The limits are shown from top to bottom for $m_H = 200$ GeV, $m_H = 300$ GeV, $m_H = 400$ GeV and $m_H = 500$ GeV.

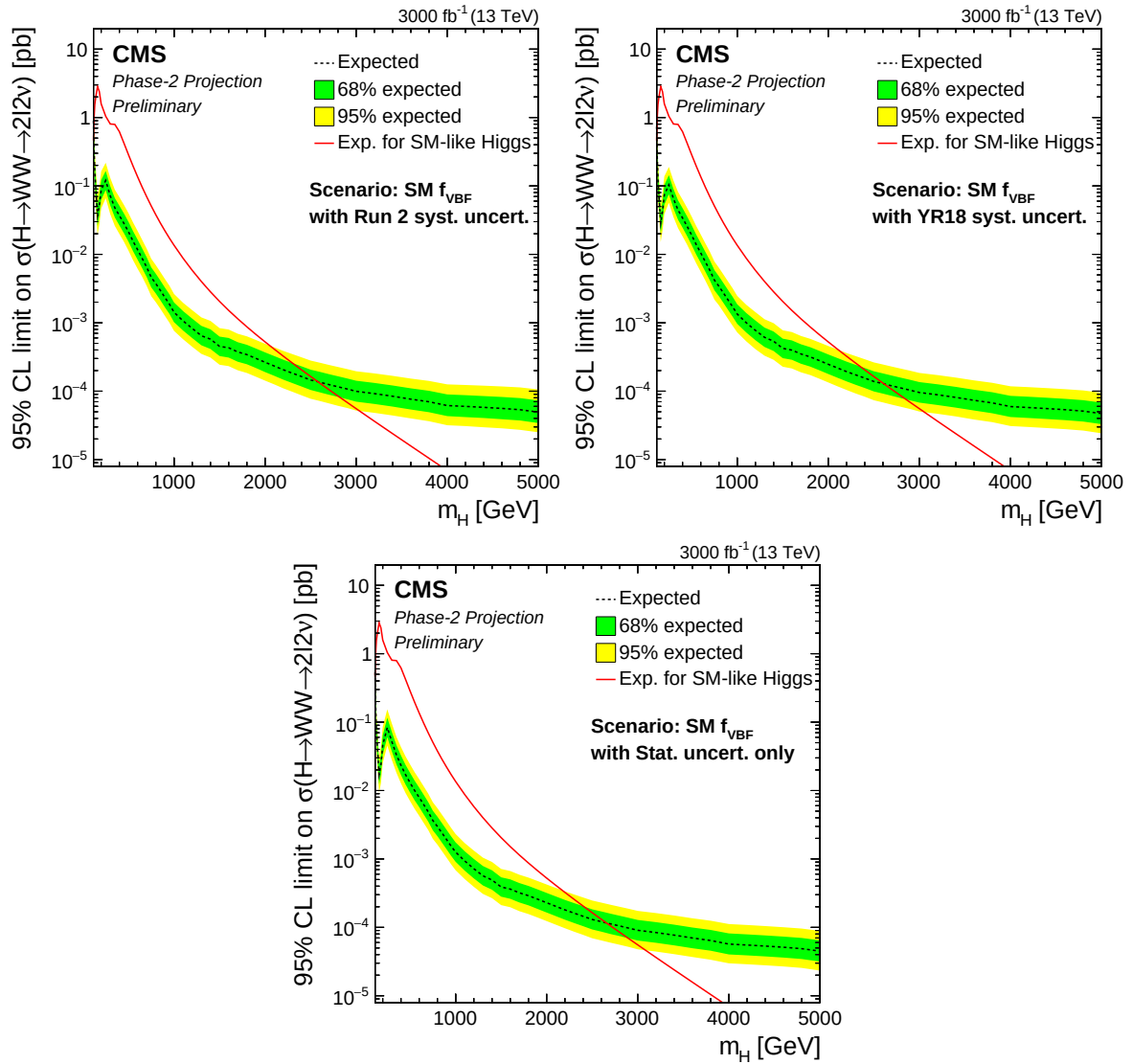


Figure 9: Phase-2 extrapolated limits for the SM VBF fraction scenario, using the Run 2 uncertainties scenario (top left), the YR18 uncertainties scenario (top right) and the statistical uncertainties only scenario (bottom).

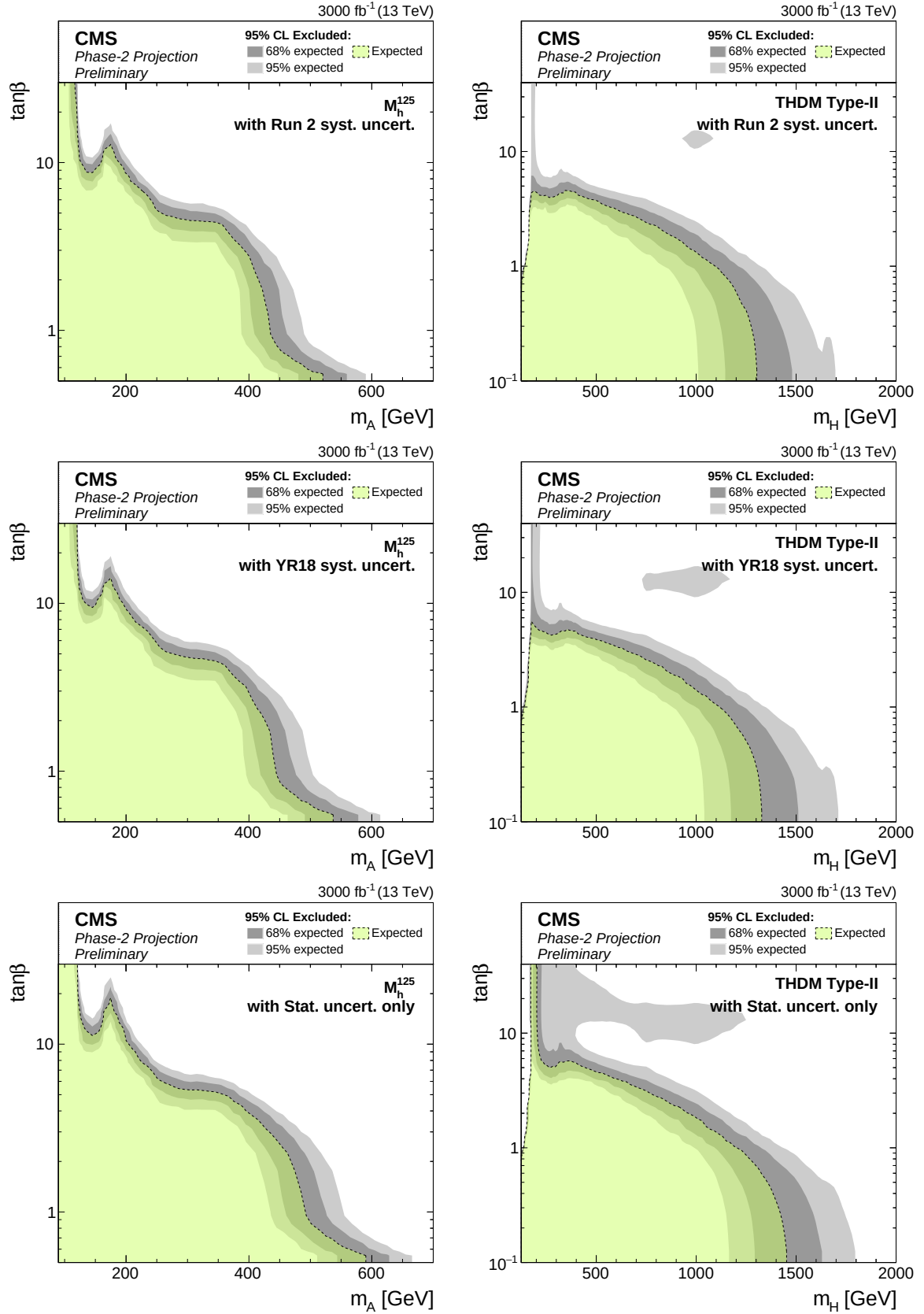


Figure 10: Phase-2 extrapolated limits for the MSSM M_h^{125} scenario (left) and a Type-II THDM over the m_H - $\tan\beta$ -plane (right). The limits are shown using the Run 2 uncertainties scenario (top), the YR18 uncertainties scenario (middle) and the statistical uncertainties only scenario (bottom).

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